Basic Power bi Assignment

Question 1: What is Power BI and why is it used in businesses?

Answer:

Power BI is a powerful Business Intelligence and Data Visualization tool developed by Microsoft. It allows users to connect to various data sources, transform raw data into meaningful insights, and create interactive reports and dashboards.

Business Use: It is used in businesses to make data-driven decisions, track Key Performance Indicators (KPIs) in real-time, and identify market trends to improve overall performance.

Question 2: Name and explain the three main components of Power BI.

Answer:

Power BI Desktop: A free Windows application used for data modeling, data transformation (using Power Query), and creating reports.

Power BI Service (SaaS): A cloud-based online service used for publishing reports, creating dashboards, and sharing insights with team members.

Power BI Mobile: Mobile applications for Windows, iOS, and Android devices that allow users to view and interact with reports on the go.

Question 3: Explain the Power BI workflow.

Answer:

The standard Power BI workflow follows these steps:

Step 1: Get Data: Import data into Power BI Desktop from various sources.

Step 2: Transform Data: Clean and shape the data using the Power Query Editor.

Step 3: Modeling: Create relationships between different tables and define calculations (DAX).

Step 4: Visualization: Build interactive reports with charts and maps in Report View.

Step 5: Publish & Share: Upload the report to the Power BI Service and share it with others

Question 4: List any four data cleaning tasks that can be performed in Power Query.

Answer:

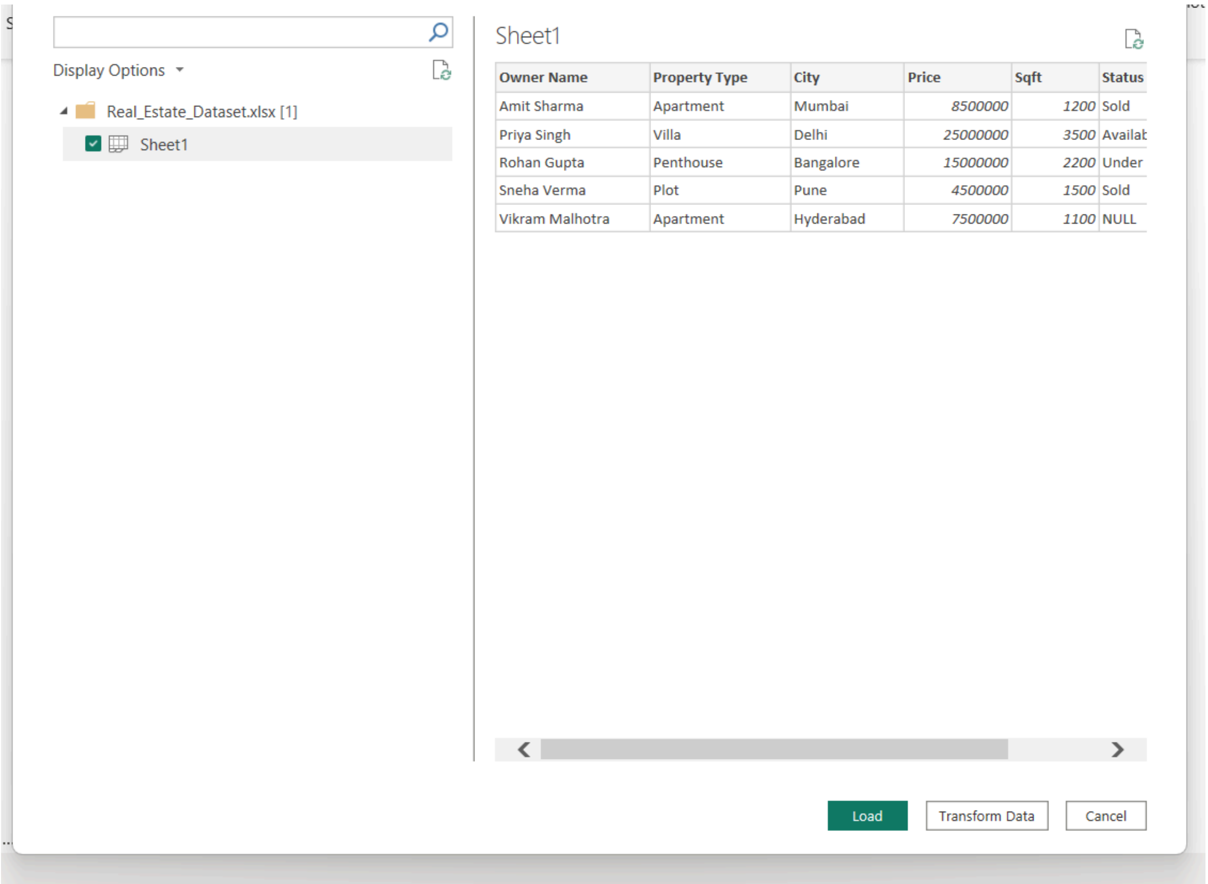
Removing Duplicates/Nulls: Deleting redundant records or empty rows from the dataset.

Changing Data Types: Converting text columns into numbers, dates, or currency formats.

Splitting Columns: Dividing a single column into multiple ones (e.g., Splitting "Full Name" into "First" and "Last Name").

Replacing Values: Finding specific errors (like "NaN" or "unknown") and replacing them with correct values.

Question 5: Write step by step instruction to load the above dataset.



To load the Real_Estate_Dataset into Power BI, follow these steps:

Step 1: Open Power BI Desktop on your computer.

Step 2: Click on the "Get Data" icon under the Home tab in the top ribbon.

Step 3: Select "Excel workbook" from the list of common data sources.

Step 4: Browse your folders, select the file named "Real_Estate_Dataset.xlsx", and click Open.

Step 5: In the Navigator window that appears, check the box next to the table or sheet you want to load (e.g., "Sheet1").

Step 6: Click on the "Transform Data" button to open the Power Query Editor for any necessary cleaning, or click "Load" to bring the data directly into your report.

Question 6: Define Data View, Report View, and Model View.

Answer:

Report View: The main canvas where you design visuals, add slicers, and create the layout of your report.

Data View: A view where you can see the actual tables and data in your model. This is where you can create new columns and measures.
Model View: A view that shows the relationship between your tables. It is used to manage complex data schemas and connections.

Question 7: Discuss the different data sources that Power BI supports.

Answer:

Power BI supports over 100 different data sources, including:

Files: Excel (.xlsx), CSV, Text, XML, JSON, and PDF.

Databases: SQL Server, Oracle, MySQL, and PostgreSQL.

Online Services: Google Analytics, Salesforce, and SharePoint.

Azure/Cloud: Azure SQL Database and Azure Blob Storage.

Question 8: Split Owner Name to create two new column as first name and last name.

Before Splitting:

```
= Table.TransformColumnTypes(#"Promoted Headers",{{"Owner Name", type text}, {"Property Type", type text}, {"City",
```

	Owner Name	Property Type	City	Price	Sqft	Status
1	Amit Sharma	Apartment	Mumbai	8500000	1200	Sold
2	Priya Singh	Villa	Delhi	25000000	3500	Available
3	Rohan Gupta	Penthouse	Bangalore	15000000	2200	Under Construct
4	Sneha Verma	Plot	Pune	4500000	1500	Sold
5	Vikram Malhotra	Apartment	Hyderabad	7500000	1100	NULL

After Splitting:

```
= Table.RenameColumns(#"Changed Type1",{{"Owner Name.1", "First Name"}, {"Owner Name.2", "Last Name"}})
```

	First Name	Last Name	Property Type	City	Price	Sqft
1	Amit	Sharma	Apartment	Mumbai	8500000	
2	Priya	Singh	Villa	Delhi	25000000	
3	Rohan	Gupta	Penthouse	Bangalore	15000000	
4	Sneha	Verma	Plot	Pune	4500000	
5	Vikram	Malhotra	Apartment	Hyderabad	7500000	